

SM PHARMACEUTICALS SDN. BHD.

FUSIX (FRUSEMIDE) TABLET 40MG

Description:

White to off white, round, flat tablet having a break line on one side.

Composition:

Each tablet contains: Frusemide 40 mg

Actions and mode or mechanisms of action:

Pharmacodynamics:

The pharmacologic effects of frusemide are similar to those of ethacrynic acid. The exact mode of action of frusemide has not been clearly defined. In contrast to ethacrynic acid, it does not bind sulfhydryl groups of renal cellular proteins. Frusemide inhibits the reabsorption of electrolytes in the ascending limb of the loop of Henle. The drug also decreases reabsorption of sodium and chloride and increases potassium excretion in the distal renal tubule and exerts a direct effect on electrolyte transport at the proximal tubule. Frusemide does not inhibit carbonic anhydrase and is not an aldosterone antagonist.

Frusemide diuresis results in enhanced excretion of sodium, chloride, potassium, hydrogen, calcium, magnesium, ammonium, bicarbonate, and possibly phosphate. Chloride excretion exceeds that of sodium. In studies in patients with normal renal function, the diuretic response was similar following oral or IV administration of equal doses of frusemide than after equal oral doses. Excessive losses of potassium, hydrogen, and chloride may result in metabolic alkalosis. Urinary pH usually falls after administration of frusemide. However, increased bicarbonate excretion in some patients may temporarily raise urinary pH. Low doses of frusemide promote uric acid retention while large IV doses may cause temporary uricosuria. Maximum diuresis and electrolyte loss is greater with frusemide than with the thiazides or most other diuretics except ethacrynic acid. Like the thiazide diuretics and ethacrynic acid, the effectiveness of frusemide is independent of the acid-base balance of the patient.

Frusemide has some renal vasodilator effect. Renal vascular resistance decreases and renal blood flow increases following administration of the drug. When large doses of frusemide are administered to patients with chronic renal insufficiency, glomerular filtration rate may be increased temporarily. A fall in renal blood flow and glomerular filtration rate may occur if excessive drug-induced diuresis results in a reduction in plasma volume.

As with other diuretics, a hypotension effect may result from decreased plasma volume in patients receiving frusemide. However, the drug has been reported to produce only mild decreases in the supine systolic blood pressure and in the erect systolic and diastolic blood pressures when administered alone in the recommended oral dosage.

Frusemide appears to have less effect on carbohydrate metabolism and blood glucose concentrations than do the thiazides. However, the drug may cause elevations of blood glucose, glycosuria, and alterations in glucose tolerance possibly as a result of hypokalemia.

Pharmacokinetics:

Absorption:

In one study in patients with normal renal function, approximately 60% of a single 80-mg oral dose of frusemide was absorbed from GI tract. When administered to fasting adults in this dosage, the drug appeared in the serum within 10 minutes, reached a peak concentration of 2.3 µg / ml in 60-70 minutes, and was almost completely cleared from the serum in 4 hours. When the same dose was given after a meal, the serum concentration of frusemide increased slowly to a peak of about 1 µg / ml after 2 hours and similar concentrations were present 4 hours after ingestion. However, a similar diuretic response occurred regardless of whether the drug was given with food or to fasting patients. In another study, the rate and extent of absorption varied considerably when 1 g of frusemide was given orally to uremic patients. An average of 76% of a dose was absorbed, and peak plasma concentrations were achieved within 2-9 hours (average 4.4 hours). Serum concentrations required to produce maximum diuresis are not known, and it has been reported that the magnitude of response does not correlate with either the peak or the mean serum concentrations.

The diuretic effect of orally administered frusemide is apparent within 30 minutes to 1 hour and is maximal in the first or second hour. The duration of action is usually 6-8 hours. The maximum hypotension effect may not be apparent until several days after frusemide therapy is begun. In patients with severely impaired renal function, the diuretic response may be prolonged.

Distribution

Only limited information is available on the distribution of frusemide. The drug crosses the placenta and is distributed into milk.

Frusemide is approximately 95% bound to plasma proteins in both normal and azotemic patients.

Elimination

Plasma concentrations of frusemide decline in a biphasic manner. Various investigators have reported a wide range of elimination half-lives for frusemide. The elimination half-life was more prolonged in 1 patient with concomitant liver disease.

In patients with normal renal function, a small amount of frusemide is metabolized in the liver to the defurfurylated derivative, 4-chloro-5-sulfamoylanthranilic acid. Frusemide and its metabolite are rapidly excreted in urine by glomerular filtration and by secretion from the proximal tubule. In patients with normal renal function, approximately 50% of an oral dose is excreted in the urine within 24 hours. 69-97% of these amounts is excreted in the first 4 hours. The remainder of the drug is eliminated by nonrenal mechanisms including degradation in the liver and excretion of unchanged drug in the feces. In patients with marked renal impairment without liver disease, nonrenal clearance of frusemide is increased so that up to 98% of the drug is removed from the plasma within 24 hours. Frusemide is not removed by hemodialysis.

Indications:

Frusemide is indicated in the treatment of edema associated with congestive heart failure, hepatic cirrhosis and renal disease (including nephrotic syndrome). It is also indicated as an adjunct in the treatment of acute pulmonary edema and hypertensive crisis.

Contraindications:

Frusemide is contraindicated in patients with anuria. The drug contraindicated for further use if increasing azotemia and/or oliguria occur during the treatment of severe, progressive renal disease. In patients with hepatic coma or electrolyte depletion, therapy should not be instituted until the basic condition is improved or corrected. Frusemide is also contraindicated in patients with a history of hypersensitivity to the drug and to sulphonamides.

Side Effects / Adverse Reactions:

The most common side-effect associated with frusemide therapy is fluid and electrolyte imbalance including hyponatraemia, hypokalaemia, and hypochloraemic alkalosis, particularly after large doses or prolonged administration. Unlike the thiazide diuretics, it increases the urinary excretion of calcium. Nephrocalcinosis has been reported when frusemide has been used to treat preterm infants. Frusemide may provoke hyperglycaemia and glycosuria, but probably to a lesser extent than the thiazide diuretics. It may cause hyperuricaemia and precipitate attacks of gout in some patients.

Other side-effects are relatively uncommon, and include gastro-intestinal disturbances, blurred vision, yellow vision, dizziness, headache, and orthostatic hypotension. Skin rashes and photosensitivity reactions, although rare, may be severe and hypersensitivity reactions including interstitial nephritis occur rarely. Pancreatitis is more common at high doses and cholestatic jaundice has been reported. Bone marrow depression may occur rarely: agranulocytosis, thrombocytopenia, and leucopenia have been reported.

Tinnitus and deafness may rarely occur in particular during rapid high dose parenteral frusemide therapy. Rarely, deafness may be permanent particularly if frusemide has been given to patients taking other ototoxic drugs.

Diuresis (strong) – The most commonly encountered adverse effects attributable to frusemide are fluid depletion and electrolyte disturbance arising from frusemide's diuretic actions.

Levels of serum lipids – Most studies into the effects of diuretics on bloods-lipids concentrations have used thiazides and long-term studies have suggested that adverse effects on blood lipids may be transitory, although the complete lipid-lipoprotein analysis necessary for full evaluation of the effects has generally not been preformed. The few studies into the effects of frusemide suggest that it may adversely influence blood-lipid concentrations during short-term administration.

Adverse dermatologic and / or hypersensitivity reactions to frusemide include purpura, photosensitivity, rash, urticaria, pruritus, exfoliative dermatitis, erythema multiforme, and necrotizing angitis (vasculitis, cutaneous vasculitis). Patients with known sulfonamide

sensitivity may show allergic reactions to frusemide. Anaphylaxis, manifested as urticaria, angiodema, and hypotension, occurred with 5 minutes after IV administration of frusemide in at least one patients; subsequent intradermal skin testing showed sensitivity to frusemide and other sulfonamides.

Precautions / Warnings:

Although frusemide is used in high doses for oliguria due to chronic or acute renal insufficiency it should not be given in anuria or in renal failure due to nephrotoxic or hepatotoxic drugs nor in renal failure associated with hepatic coma. Frusemide should not be given in pre-comatose states associated with hepatic cirrhosis. It should be used with care in patients with prostatic hypertrophy or impairment of micturition since it can precipitate acute urinary retention.

Frusemide should be used with caution in pregnancy.

Lactation – Frusemide passes into breast milk, and inhibit lactation. Therefore, it must not be given during breast feeding.

Drug Interactions

Frusemide may enhance the nephrotoxicity of cephalosporin antibiotics such as cephalothin. It can enhance the ototoxicity of aminoglycoside antibiotics. Like other diuretics, frusemide enhances the hypotensive action of antihypertensive drugs. Particular care should be taken with ACE inhibitors since combination with frusemide can result in marked reduction in blood pressure.

Cisplatin – The concomitant administration of cisplatin and frusemide carries the risk of inducing hearing defects, the 2 drugs should not be used simultaneously.

Steroids – When a glucocorticoid is administered during diuretic treatment the potassium-lowering effect of the steroid should be borne in mind.

Lithium – Concurrent use with loop diuretics may provoke lithium toxicity because of reduced renal clearance and is not recommended unless patient can be closely monitored.

Phenytoin – Phenytoin significantly reduces absorption, plasma levels and renal response of frusemide.

NSAIDs – May antagonize the natriuresis and increase in plasma renin activity (PRA) caused by loop diuretics; indomethacin, and possibly other NSAIDs with the exception of diflunisal, may also reduce the increase in urine volume caused by loop diuretics, possibly by inhibiting renal prostaglandin synthesis and / or by causing sodium and fluid retention.

In addition, concurrent use of NSAIDs with a diuretic may increase the risk of renal failure secondary to a decrease in renal blood flow caused by inhibition of renal prostaglandin synthesis. In the premature neonate, administration of 1 mg / kg of frusemide immediately following indomethacin has been shown to prevent or reduce indomethacin-reduced adverse renal effects without interfering with ductus arteriosus closure.

Probenecid – Probenecid has been found to increase serum concentrations of frusemide by inhibiting active renal tubular secretion.

Recommended Dosage, Dosage Schedule and Route of Administration:

In the treatment of edema, the usual initial dose is 40 mg once daily by mouth, adjusted as necessary according to response. Mild cases may respond to 20 mg daily or 40 mg on alternate days. Some patients may require doses of 80 mg or more daily given one or two doses daily, or intermittently. Severe cases may require gradual titration of frusemide dosage up to 600 mg daily. If necessary further doses may be given, increasing by 20 mg increments and not given more often than every 2 hours.

For children, the usual dose by mouth is 1 to 3 mg per kg body-weight daily up to a maximum of 40 mg daily. In the treatment of hypertension, frusemide is given in doses of 40 to 80 mg daily, either alone, or in conjunction with other antihypertensive agents.

High-dose therapy.

Patients who do not respond to a dose of 1 g probably require dialysis. If the response to either method of administration is satisfactory, the effective dose (of up to 1 g) may then be repeated every 24 hours. Dosage adjustments should subsequently be made according to the patient's response. Alternatively, treatment may be maintained by mouth; 500 mg should be given by mouth for each 250 mg required by injection.

In the treatment of chronic renal insufficiency, an initial dose of 250 mg may be given by mouth, increased, if necessary in steps of 250 mg every 4 to 6 hours to a maximum of 1.5 g in 24 hours; in exceptional cases up to 2.0 g in 24 hours, may be given.

Dosage adjustments should subsequently be made according to the patient's response.

During treatment with these high-dose forms of frusemide therapy, careful laboratory control is essential. Fluid balance and electrolytes should be carefully controlled and, in particular, in patients with shock, measures should be taken to correct the blood pressure and circulating blood volume, before commencing this type of treatment. High dose frusemide therapy is contraindicated in renal failure caused by nephrotoxic or hepatotoxic agents, and in renal failure associated with hepatic coma.

Symptoms and Treatment for Overdose and Antidote(s):

Overdose will lead to fluid and salt depletion subsequent to initial excessive diuresis. Medical help must be called in without delay so as to ensure immediate measures for an adequate restoration of water and electrolytes balances.

Storage Conditions, User Instructions and Pharmaceutical Precautions:

KEEP ALL MEDICINES OUT OF REACH OF CHILDREN

Store below 30°C .

Protect from direct light and heat.

Packing / Pack Sizes:

Plastic containers of 30's ,100's and 500's (For Export Only)

Blister-packs of 10's, 10x10's/Box, 50x10's/Box and 100x10's/Box.

Shelf life: 3 years

Manufacturer and Product Registration Holder:

SM PHARMACEUTICALS SDN BHD (218620-M)

Lot 88, Sungai Petani Industrial Estate

08000 Sungai Petani,

Kedah Darul Aman,

Malaysia.

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